JANHAVI PATIL

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EDUCATION

University at Albany, New York 2024-2026
Masters in Data Science

Vishwakarma Institute of Information Technology, Pune 2020-2024
Bachelor of Artificial Intelliegnce & Data Science (Honors in Digital Marketing with ML Applications)

SKILLS

Languages & Databases: Python, SQL, Java, MySQL, PostgreSQL
AI/ML & Data Science: Machine Learning, NLP, Time Series Forecasting (ARIMA), Feature Engineering, Model Evaluation, Statistical Analysis
Model Evaluation LLM & Generative AI: Retrieval-Augmented Generation (RAG), Prompt Engineering, OpenAI API, LangChain, Vector Databases (FAISS)
Cloud & Tools: AWS (S3, EC2), Microsoft Azure, Docker (basic), Git, GitHub, Linux
Data Analysis & Visualization: Pandas, NumPy, EDA, Data Visualization, Matplotlib/Seaborn
Web/Backend (Basic): REST APIs, Spring Boot

EXPERIENCE

AI/ML Intern — Manufacturing Analyst, Prabha Industries Jan 2024-Jul 2024

- Analyzed production data to support AI/ML initiatives across manufacturing operations, helping improve operational efficiency.
- Developed predictive models for defect detection, enabling earlier identification of quality issues in production output.
- Built interactive dashboards surfacing key production metrics, supporting data-driven decision-making for manufacturing stakeholders.

PROJECTS & PUBLICATIONS

Manufacturing Incident Investigation & Root Cause Analysis System | *Live Demo* | *GitHub* Aug 2026

- Built and deployed a Streamlit-based (Python, Plotly) manufacturing incident investigation system analyzing sensor data (temperature, vibration, pressure, motor current, RPM), maintenance history, and incident records to identify root causes.
- Architected a modular analysis pipeline sensor anomaly analysis, maintenance correlation, incident investigation, and risk scoring to flag abnormal conditions and recommend corrective actions.
- Implemented flexible CSV/Excel data ingestion with automatic and manual column mapping to handle varying dataset schemas.

Player Performance Dashboard — FIFA 2026 | *Tableau* | *Live Dashboard* Aug 2026

- Built and published an interactive Tableau dashboard visualizing and comparing FIFA 2026 player performance across attributes, positions, and match statistics.
- Designed drill-down views and KPI cards to surface player and team performance trends for scouting- and fan-analytics-style insights.

AI-Powered Air Quality Forecasting & Analysis System | *Python, Machine Learning, Time Series* Mar 2026

- Developed an XGBoost/LSTM-based AQI forecasting model for Kolhapur, benchmarked against a baseline ARIMA model using historical pollutant data.
- Engineered features from multi-dimensional pollutant data (PM2.5, PM10, NO_2, SO_2, CO) to capture temporal and seasonal pollution patterns.
- Cleaned and preprocessed sensor data — handling missing values, outliers, and inconsistencies — improving data reliability ahead of model training.

LLM-Powered Retrieval-Augmented Generation (RAG) System | *Python, FAISS, OpenAI, LangChain* Jan 2026

- Designed and implemented an end-to-end RAG pipeline integrating document retrieval with LLM-based response generation.
- Built semantic search using vector embeddings and FAISS for efficient similarity-based document retrieval.
- Implemented document chunking, embedding generation, and context-aware prompt engineering to improve response accuracy.
- Optimized chunking strategy and embedding batch size to reduce query latency and support enterprise-scale document volumes.

Real-Time NLP-Based News Classification Pipeline | *Python, Scikit-learn* | *Paper Link* Jan 2025

- Designed an end-to-end NLP pipeline classifying real-time news articles into structured categories using tokenization, stopword removal, and TF-IDF vectorization.
- Trained and optimized a Naive Bayes classifier for scalable text classification, evaluated using accuracy, precision, recall, and F1-score.

Smart Plant Monitoring System | *IoT, Data Analytics* | *Paper Link* Jan 2025

- Built sensor-based automated irrigation system enabling real-time monitoring and efficient water usage.